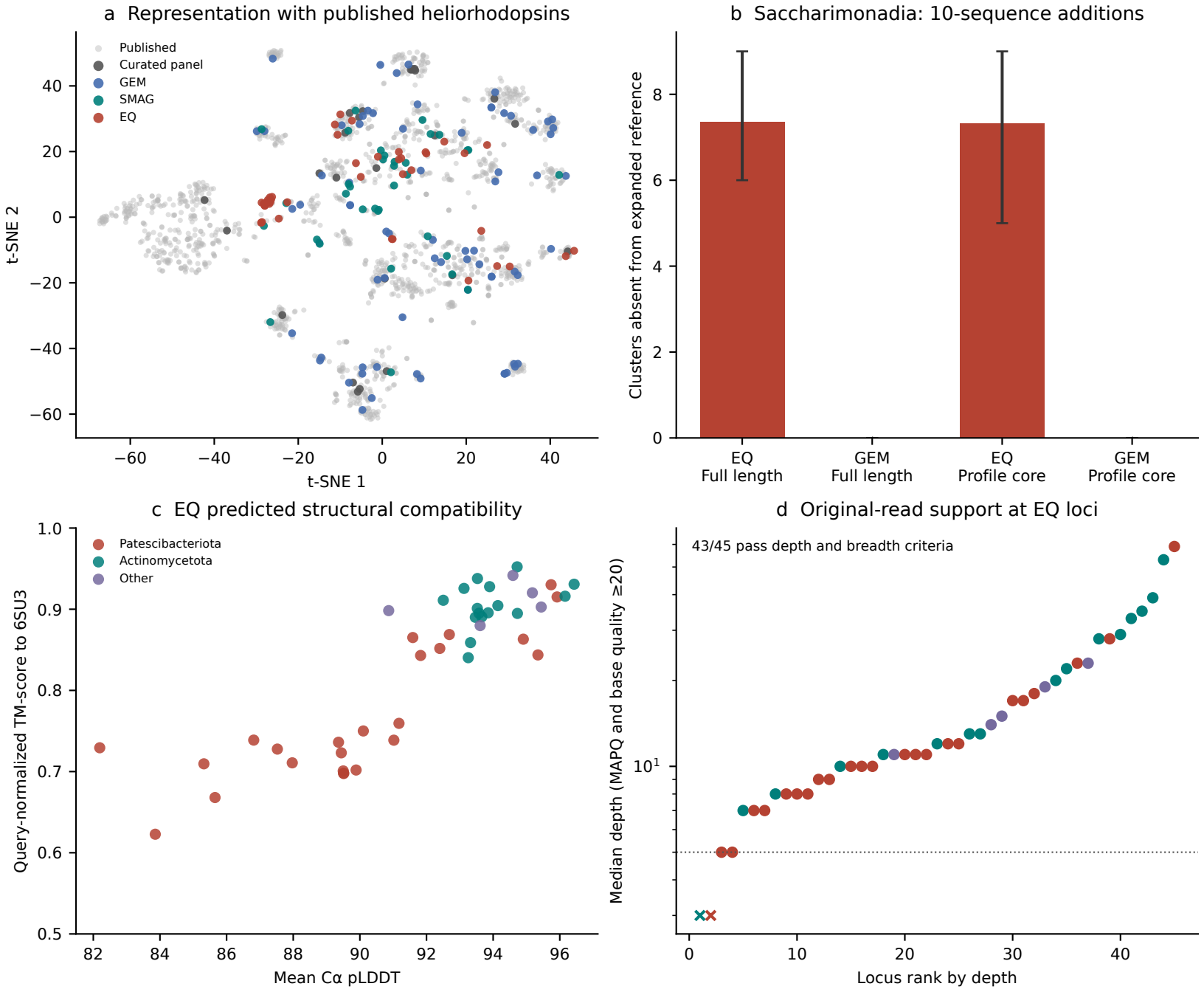

a Representation with published heliorhodopsins
Published
Curated panel
GEM
SMAG
EQ
t-SNE 1
t-SNE 2
b Saccharimonadia: 10-sequence additions
Clusters absent from expanded reference
EQ Full length
GEM Full length
EQ Profile core
GEM Profile core
c EQ predicted structural compatibility
Patescibacteriota
Actinomycetota
Other
Query-normalized TM-score to 6SU3
Mean Cα pLDDT
d Original-read support at EQ loci
43/45 pass depth and breadth criteria
Median depth (MAPQ and base quality ≥20)
Locus rank by depth